1. If $f(x) = (\cosh x)^2$, then $f'(\ln 2) =$

(a) $\frac{5}{16}$

(b) $\frac{15}{24}$

(c) $\frac{15}{4}$

(d) $\frac{3}{8}$

(e) $\frac{15}{8}$

Not Required in EXAM II (092)

2. $\frac{d}{dt}\left(\frac{2t-1}{3t+2}\right)^8 =$

(a) $\frac{56(2t-1)^7}{(3t+2)^{18}}$

(b) $\frac{72(2t-1)^7}{(3t+2)^9}$

(c) $\frac{56(2t-1)^7}{(3t+2)^9}$

(d) $\frac{24(2t-1)^7}{(3t+2)^9}$

(e) $\frac{48(2t-1)^7}{(3t+2)^{18}}$

$$8\left(\frac{2t-1}{3t+2}\right)^7 \frac{(3t+2)(2)-(2t-1)(3)}{(3t+2)^2}$$

$$= 8\,\frac{(2t-1)^7}{(3t+2)^7}\,\frac{7}{(3t+2)^2}$$

$$= \frac{56(2t-1)^7}{(3t+2)^9}$$

3. The slope of the tangent line to the graph of $f(x) = \dfrac{x^2 - 2\sqrt{x}}{2x+3}$ at $x = 1$ is equal to

$$f'(x) = \frac{(2x+3)\left(2x - \frac{1}{\sqrt{x}}\right) - (x^2 - 2\sqrt{x})(2)}{(2x+3)^2}$$

$$f'(1) = \frac{(5)(1) - (-1)(2)}{5^2}$$

$$= \frac{7}{25}$$

(a) $\frac{3}{25}$

(b) $-\frac{7}{25}$

(c) $\frac{7}{25}$

(d) $\frac{9}{25}$

(e) $-\frac{3}{25}$

4. If the position of a particle is given by the equation

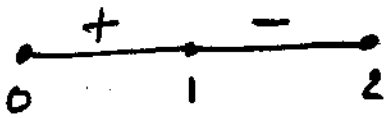
$$S(t) = 2t^3 - 9t^2 + 12t,$$

where t is measured in seconds and S in meters, then the total distance traveled by the particle during the time interval $[0, 2]$ is

$$v = S'(t) = 6t^2 - 18t + 12$$

$$= 6(t^2 - 3t + 2)$$

$$= 6(t-1)(t-2)$$



The required distance

$$= |S(1) - S(0)| + |S(2) - S(1)|$$

$$= |5 - 0| + |16 - 36 + 24 - 2 + 9 - 12|$$

$$= 5 + 1 = 6$$

(a) 1 meter

(b) 4 meters

(c) 9 meters

(d) 6 meters

(e) 5 meters

5. $\lim_{x\to 0} \dfrac{3\tan 2x - 5\tan 3x}{7x\cos x + 4\sin 5x} =$ (0/0 type)

Divide the numerator and the denominator of the expression inside the limit by $x \Rightarrow$

$$\lim_{x\to 0} \frac{\frac{3\tan 2x}{x} - \frac{5\tan 3x}{x}}{7\cos x + 4\frac{\sin 5x}{x}}$$

$$= \frac{(3)(2) - (5)(3)}{7 + (4)(5)} = \frac{6-15}{7+20}$$

$$= \frac{-9}{27} = -\frac{1}{3}$$

(a) $\frac{1}{27}$

(b) $-\frac{1}{3}$

(c) $\frac{5}{27}$

(d) $-\frac{8}{9}$

(e) $-\frac{1}{9}$

6. Which one of the following statements is **FALSE** about the function $f(x) = x^{2/3}$?

$$f'(x) = \frac{2}{3}x^{-1/3} = \frac{2}{3\sqrt[3]{x}}$$

(a) the domain of f' is $(-\infty, 0) \cup (0, \infty)$

(b) f has no horizontal tangents

(c) f is continuous at $(0, 0)$

(d) $\lim_{x\to 0^-} f'(x) = \lim_{x\to 0^+} f'(x) = +\infty$

$\lim_{x\to 0^-} f'(x) = -\infty$

$\lim_{x\to 0^+} f'(x) = +\infty$

(e) f has a vertical tangent at $x = 0$

7. If $f(x) = \cos x$, then $f^{(99)}(x) =$

$f'(x) = -\sin x$, $f''(x) = -\cos x$

$f'''(x) = \sin x$, $f^{(4)}(x) = \cos x$

$f^{(5)}(x) = -\sin x$, $f^{(6)}(x) = -\cos x$

$f^{(2m-1)}(x) = (-1)^m \sin x$, $f^{(2m)}(x) = (-1)^m \cos x$

$99 = 2(50) - 1 \Rightarrow f^{(99)}(x) = (-1)^{50} \sin x = \sin x$

(a) $\cos x$

(b) $-\cos x$

~~(c)~~ $\sin x$

(d) $-\sin x$

(e) none of the other given answers

8. If $g(x) = \sec(x^3)\ \cot(x^3)$, then $g'(x) =$

Simplify first:

$g(x) = \frac{1}{\cos(x^3)} \cdot \frac{\cos(x^3)}{\sin(x^3)}$

$= \frac{1}{\sin(x^3)} = \csc(x^3) \Rightarrow$

$g'(x) = -3x^2 \csc(x^3) \cot(x^3)$.

(a) $-3x^2 \csc(x^3)\ \tan(x^3)$

(b) $-3x^2 \sec(x^3)\ \tan(x^3)$

(c) $-3x^2 \sec(x^3)\ \cot(x^3)$

~~(d)~~ $-3x^2 \csc(x^3)\ \cot(x^3)$

(e) $-3x^2 \sec(x^3)\ \csc^2(x^3)\ \tan(x^3)$

9. If $y = x\cos^{-1}\left(\frac{x}{2}\right) - \sqrt{4-x^2}$, then $\frac{dy}{dx} =$

(a) $\frac{1}{4}\cos^{-1}\left(\frac{x}{2}\right)$

(b) $2\cos^{-1}\left(\frac{x}{2}\right) - \frac{2x}{\sqrt{4-x^2}}$

(c) $\frac{1}{4}\cos^{-1}\left(\frac{x}{2}\right) - \frac{x}{\sqrt{4-x^2}}$

(d) $\cos^{-1}\left(\frac{x}{2}\right) - \frac{2x}{\sqrt{4-x^2}}$

~~(e)~~ $\cos^{-1}\left(\frac{x}{2}\right)$

$y' = \cos^{-1}\left(\frac{x}{2}\right) - \frac{\frac{1}{2}x}{\sqrt{1-\frac{x^2}{4}}} - \frac{(-2x)}{2\sqrt{4-x^2}}$

$= \cos^{-1}\left(\frac{x}{2}\right) - \frac{x}{\sqrt{4-x^2}} + \frac{x}{\sqrt{4-x^2}}$

$= \cos^{-1}\left(\frac{x}{2}\right)$

10. $\lim_{\theta\to 0}\frac{\cos\theta - 1}{3\tan^2\theta} =$

(a) $-\frac{3}{2}$

~~(b)~~ $-\frac{1}{6}$

(c) 0

(d) $\frac{2}{3}$

(e) does not exist

It is $\frac{0}{0}$ type

Use the identity $\cos 2\alpha = 1 - 2\sin^2\alpha \Rightarrow$

$1 - \cos 2\alpha = -2\sin^2\alpha \Rightarrow$

$\lim_{\theta\to 0}\frac{\cos\theta - 1}{3\tan^2\theta} = \lim_{\theta\to 0}\frac{-2\sin^2\theta/2}{3\tan^2\theta}$

$= -\frac{2}{3}\lim_{\theta\to 0}\left(\frac{\sin\theta/2}{\tan\theta}\right)^2$

$= -\frac{2}{3}\left(\frac{1}{2}\right)^2 = -\frac{2}{3}\cdot\frac{1}{4} = -\frac{1}{6}$

Recall $\lim_{\theta\to 0}\frac{\sin m\theta}{\tan n\theta} = \frac{m}{n}$

11. If $f(x) = 3x \sin 2x$, then $\lim_{h\to 0} \frac{1}{h}\left[f\left(\frac{\pi}{2}+h\right) - f\left(\frac{\pi}{2}\right)\right] =$

$= \lim_{h\to 0} \frac{1}{h}\left[3\left(\frac{\pi}{2}+h\right)\sin(\pi+2h) - 0\right]$

(a) -3π

(b) $-3-3\pi$

(c) $-\frac{3\pi\sqrt{2}}{2}$

(d) $\frac{\sqrt{2}}{2}+3\pi$

(e) $3+6\pi$

$= \lim_{h\to 0} 3\left(\frac{\pi}{2}+h\right)\left(\frac{-\sin(2h)}{h}\right)$

$= 3\left(\frac{\pi}{2}\right)(-2) = -3\pi.$

12. If $y = x^{1/x}$, then $\frac{dy}{dx} =$

(a) $x^{-2+\frac{1}{x}}(1-\ln x)$

(b) $-x^{-3+\frac{1}{x}}\ln x$

(c) $x^{-2+\frac{1}{x^2}}(1-\ln x)$

(d) $-x^{-3+\frac{1}{x}}$

(e) $x^{-2+\frac{1}{x^2}}(1+\ln x)$

$y = x^{1/x} \Rightarrow$

$\ln y = \frac{1}{x}\ln x \Rightarrow$

$\frac{y'}{y} = \frac{1}{x}\left(\frac{1}{x}\right) - \frac{1}{x^2}\ln x \Rightarrow$

$y' = x^{1/x}\left[\frac{1}{x^2} - \frac{1}{x^2}\ln x\right]$

$= x^{1/x}\cdot x^{-2}\left[1-\ln x\right]$

$= x^{-2+1/x}(1-\ln x)$

13. If $4x^2 + y^2 = 4$, then $y'' =$

(a) $-\frac{1}{4y^3}$

(b) $\frac{16x}{y^4}$

(c) $-\frac{16}{y^3}$

(d) $-\frac{16x^3}{y^3}$

(e) $\frac{4x}{y^4}$

$\Rightarrow \frac{d}{dx}(4x^2+y^2) = \frac{d}{dx}(4) \Rightarrow$

$8x + 2yy' = 0 \Rightarrow y' = -\frac{4x}{y} \Rightarrow$

$y'' = -4\left[\frac{y - xy'}{y^2}\right] = -\frac{4}{y^2}\left[y - x\left(-\frac{4x}{y}\right)\right]$

$= -\frac{4}{y^2}\,\frac{y^2+4x^2}{y} = -\frac{4}{y^3}(4)$

$= -\frac{16}{y^3}.$

14. The equation of the horizontal tangent to the graph of $y = \sqrt{x}\,e^{-\sqrt{x}}$ is

(a) $y = \frac{\sqrt{2}}{e}$

(b) $y = \frac{1}{\sqrt{e}}$

(c) $y = \frac{1}{e}$

(d) $y = -\frac{1}{e}$

(e) $y = -\frac{1}{\sqrt{e}}$

we need to find when $y' = 0$

$y' = \sqrt{x}\left(-\frac{1}{2\sqrt{x}}e^{-\sqrt{x}}\right) + \frac{1}{2\sqrt{x}}e^{-\sqrt{x}}$

$= \frac{1}{2\sqrt{x}}e^{-\sqrt{x}}\left(-\sqrt{x}+1\right)$

$\Rightarrow y' = 0$ when $x = 1 \Rightarrow y = e^{-1}$

$\Rightarrow$ The eqn of the horizontal tangent is

$y = \frac{1}{e}.$

15. If $\cosh x = \frac{5}{3}$, $x < 0$, then the value of $9\sinh x - 20\tanh x =$

Not Required in Exam II (092)

(a) 4

(b) $-\frac{4}{15}$

(c) -24

(d) -8

(e) 12

16. If $f(x) = \ln\left[\frac{\sin^2 x \ \tan^4 x}{(1+\cos^2 x)^3}\right]$, then $f'\left(\frac{\pi}{4}\right) =$

Simplify first:

$f(x) = 2\ln|\sin x| + 4\ln|\tan x| - 3\ln(1+\cos^2 x)$

$\Rightarrow f'(x) = \frac{2\cos x}{\sin x} + \frac{4\sec^2 x}{\tan x} - \frac{3(-2\cos x \sin x)}{1+\cos^2 x}$

(a) 8

(b) 16

~~(c)~~ 12

(d) 14

(e) 10

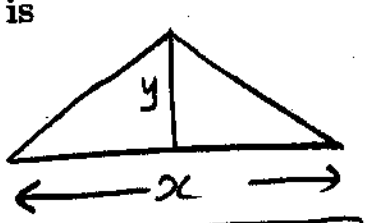
$f'\left(\frac{\pi}{4}\right) = 2\left(\frac{1/\sqrt{2}}{1/\sqrt{2}}\right) + \frac{4(2)}{1} + \frac{6(1/\sqrt{2})(1/\sqrt{2})}{1+1/2}$

$= 2 + 8 + \frac{3}{3/2} = 2+8+2 = 12$

17. The altitude of a triangle is increasing at a rate of $\frac{1}{2}$ cm/min while the area of the triangle is decreasing at a rate of $\frac{3}{2}$ cm^2/min. **The rate at which the base of the triangle is changing when the altitude is 8 cm and the area is 80 cm^2 is equal to**

Base $= x$, Altitude $= y \Rightarrow$

Area $A = \frac{1}{2} x y$



$\frac{dy}{dt} = \frac{1}{2}$, $\frac{dA}{dt} = -\frac{3}{2}$

$\Rightarrow \frac{dA}{dt} = \frac{1}{2} x \frac{dy}{dt} + \frac{1}{2} y \frac{dx}{dt}$

when $y = 8$, $x = 20 \Rightarrow$

when $y=8$, $A=80 \Rightarrow$ $80 = \frac{1}{2}(x)(8)$, $x = 20$

$-\frac{3}{2} = \frac{1}{2}(20)\left(\frac{1}{2}\right) + \frac{1}{2}(8)\frac{dx}{dt}$

$-\frac{3}{2} = 5 + 4\frac{dx}{dt} \Rightarrow 4\frac{dx}{dt} = -\frac{13}{2} \Rightarrow$

$\frac{dx}{dt} = -\frac{13}{8}$.

(a) $\frac{7}{8}$ cm/min

(b) $-\frac{15}{8}$ cm/min

(c) $\frac{11}{8}$ cm/min

~~(d)~~ $-\frac{13}{8}$ cm/min

(e) $-\frac{11}{8}$ cm/min

18. If m is the slope of the tangent line to the graph of $2^{x+y} = x^2 + xy^2 + 1$ at the point $(-1, 1)$, then the product $(2 + \ln 2)\, m$ is equal to

use implicite differentiation:

$2^{x+y}\left[1 + y'\right]\ln 2 = 2x + y^2 + 2xy'$

at $(-1,1) \Rightarrow y' = m \Rightarrow$

$2^0\left[1+m\right]\ln 2 = -2 + 1 - 2m \Rightarrow$

$\ln 2 + m\ln 2 = -1 - 2m \Rightarrow$

$m(2 + \ln 2) = -1 - \ln 2$

(a) 1

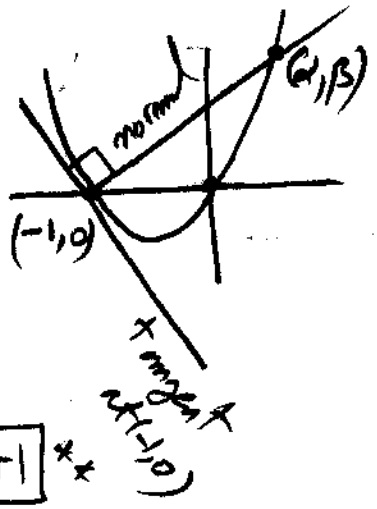
(b) $2 - \ln 2$

(c) -1

~~(d)~~ $-1 - \ln 2$

(e) $-2 + \ln 2$

19. If the normal line to the parabola $y = x^2 + x$ * at the point $(-1, 0)$ intersects the parabola a second time at the point (α, β), then $\alpha - \beta =$



slope of the tangent at $(-1,0)$ is

$y'|_{(-1,0)} = (2x+1)|_{(-1,0)} = -1$

(a) -1

(b) -3

slope of normal line $= 1 \Rightarrow$ eqn of normal $y - 0 = 1(x+1) \Rightarrow \boxed{y = x+1}$ **

(c) 1

(d) -2

(e) 2

$(*), (**) \Rightarrow x^2 + x = x + 1 \Rightarrow x^2 = 1 \Rightarrow x = -1, x = 1$

$\Rightarrow \alpha = 1 \Rightarrow \beta = 2 \Rightarrow \alpha - \beta = -1$

20. If $f(x) = \begin{cases} 3, & \text{if } x \le 0 \\ 3 - x, & \text{if } 0 < x < 2 \\ \dfrac{1}{3-x}, & \text{if } x \ge 2 \end{cases}$, then f is **not differentiable** at

(a) $x = 0$, and 2 only

(b) $x = 2$, and 3 only

(c) $x = 3$ only

(d) $x = 0$, 2, and 3

(e) $x = 0$, and 3 only

$$f'(x) = \begin{cases} 0, & x < 0 \\ 3, & 0 < x < 2 \\ \dfrac{1}{(3-x)^2}, & x > 2 \end{cases}$$

$\lim_{x \to 0^-} f'(x) = 0 \neq \lim_{x \to 0^+} f'(x) = 3$

$\Rightarrow$ Not Differentiable at 0

$\lim_{x \to 2^-} f'(x) = 3 \neq \lim_{x \to 2^+} f'(x) = \lim_{x \to 2^+} \dfrac{1}{(3-x)^2} = 1$

$\Rightarrow$ No Differentiable at 2

Also f is discontinuous at $x = 3$ since $f(x) = \dfrac{1}{3-x}$, $x \ge 2$ and f is undefined at 3

$\Rightarrow$ f is Not differentiable at 3